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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

College Course Registration and Subject Enrolment System

Submitted by

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DECLARATION

I hereby declare that the project entitled "**College Course Registration and Subject Enrolment System"** submitted in partial fulfillment of the requirements for the course **java programming** is a **record of my original work** carried out under the guidance of **MrAravind**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma or other similar title at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

AJAY KUMAR T

CERTIFICATE

This is to certify that the project entitled **"College Course Registration and Subject Enrolment System"** is a **bona fide work** carried out by **AJAY KUMAR T B.Sc CS (2025-2028),COMPUTER SCIENCE**, **KG College of Arts & Science**, in partial fulfilment of the requirements for the course **Java programming**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date:10/4/2026

Submitted for the Viva-voce Examination held on _____10/4/2026_____

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ABSTRACT

The **College Course Registration and Subject Enrolment System** is a software application designed to simplify and automate the process of course selection, registration, and enrolment in educational institutions. Traditional manual methods of course registration are time-consuming, error-prone, and inefficient, often leading to issues such as scheduling conflicts, data redundancy, and lack of transparency. This system aims to address these challenges by providing a centralized, user-friendly, and efficient digital platform for both students and administrators.The system allows students to view available courses, check eligibility criteria, and enroll in subjects based on their academic requirements. It provides real-time updates on course availability, prerequisites, and timetable schedules, helping students make informed decisions. Additionally, the system prevents common issues such as duplicate registrations and time clashes by implementing validation rules and automated checks.From the administrative perspective, the system offers tools to manage courses, allocate faculty, define course capacities, and monitor student enrolments. It also generates reports related to student registrations, course demand, and academic performance, aiding in effective decision-making and resource planning. The integration of a secure login system ensures that only authorized users can access and manage data, thereby maintaining confidentiality and data integrity. The application is typically developed using programming languages such as Java, along with database systems like MySQL for efficient data storage and retrieval. The system follows a modular design approach, making it scalable and adaptable to future enhancements, such as online fee payment, attendance tracking, and result management. The application is typically developed using programming languages such as Java, along with database systems like MySQL for efficient data storage and retrieval. The system follows a modular design approach, making it scalable and adaptable to future enhancements, such as online fee payment, attendance tracking, and result management.

1. INTRODUCTION

In today's rapidly evolving educational environment, the need for efficient and reliable academic management systems has become essential. Colleges and universities handle a large number of students, courses, and administrative tasks, making manual processes increasingly difficult to manage. Traditional methods of course registration, which often involve paperwork and long queues, are not only time-consuming but also prone to errors such as incorrect data entry, duplication, and scheduling conflicts. To overcome these limitations, the College Course Registration and Subject Enrolment System is developed as a modern digital solution.

This system is designed to automate and streamline the entire process of course registration and subject enrolment. It provides a centralized platform where students can easily access course details, check eligibility, and enroll in subjects according to their academic requirements. By offering real-time information about course availability, prerequisites, and schedules, the system helps students make better decisions and avoid conflicts in their timetable.

For administrators and faculty members, the system simplifies the management of courses, student records, and enrollment data. It enables efficient allocation of subjects, monitoring of student registrations, and generation of reports for academic planning. The system also ensures data accuracy and consistency by minimizing human intervention and providing automated validation mechanisms.

Furthermore, the use of secure login and authentication features ensures that sensitive information is protected and accessible only to authorized users. The system can be implemented using technologies such as Java for application development and databases like MySQL for data management, making it both robust and scalable.

1.1 Objective

- **To automate the registration process**
  Replace the manual method of course registration with a computerized system to save time and reduce paperwork. reduce errors and duplication

1. **To ensure efficient course management**
   Help administrators manage courses, student records, and faculty allocation effectively.
2. **To provide real-time information**
   Offer up-to-date details about course availability, seat capacity, prerequisites, and timetables.
3. **To improve data security**
   Protect student and institutional data through secure login and access control mechanisms

4. **To generate reports and analytics**
   Facilitate easy generation of reports related to student enrollment, course demand, and academic planning.
5. **To enhance overall efficiency**
   Reduce administrative workload and improve the overall performance of academic processes.
6. **To support scalability and future enhancements**
   Design the system in a way that allows adding features like online fee payment, attendance tracking, and result management in the future.

1.2 Overview

The College Course Registration and Subject Enrollment System is a digital platform developed to manage and simplify the process of course selection and student enrollment in educational institutions. It replaces traditional manual methods with an automated system that allows students to register for courses online. Through this system, students can view available subjects, check prerequisites, and select courses based on their academic needs. The platform ensures that the registration process is quick, efficient, and accessible from anywhere.

The system is designed with different modules to serve the needs of students, administrators, and faculty members. Students can log in to their accounts, choose subjects, and monitor their enrollment status. Administrators have the ability to add or remove courses, set seat limits, assign faculty, and manage student data. The system also includes validation features to prevent issues such as duplicate registrations and timetable clashes, ensuring smooth and accurate operation.

Overall, the system improves the efficiency and transparency of academic management. It reduces paperwork, minimizes errors, and saves time for both students and staff. With secure access and a well-structured database, the system maintains data integrity and confidentiality. Additionally, it provides reporting features that help institutions make better decisions regarding course planning and resource allocation, making it an essential tool for modern education systems.

1.3 Features

The key features of the College Course Registration and Subject Enrollment System are:

1. **User Authentication and Login**
   Secure login system for students, faculty, and administrators with unique usernames and passwords.

2. **Course Browsing and Selection**
   Students can view available courses, subject details, credits, and prerequisites before selecting them.

3. **Online Registration**
   Enables students to register for courses anytime and from anywhere without manual paperwork.

4. **Real-Time Course Availability**
   Displays updated information about available seats and course status.

5. **Prerequisite Checking**
   Automatically verifies whether a student meets the required conditions before enrolling in a subject.

6. **Timetable Management**
   Prevents schedule clashes by checking time conflicts between selected courses.

7. **Admin Dashboard**
   Allows administrators to add, update, or delete courses, manage student records, and assign faculty.

8. **Faculty Management**
   Helps in assigning instructors to specific subjects and managing teaching schedules.

9. **Enrollment Tracking**
   Students can view their registered courses and track enrollment status.

10. **Report Generation**
    Generates reports on student enrollment, course demand, and academic records.

11. **Data Security and Backup**
    Ensures safe storage of data with backup and recovery options.

12. **Notifications and Alerts**
    Provides updates about registration deadlines, course changes, and important announcements.

13. **Scalability**
    The system can be expanded to include additional features like fee payment, attendance, and results.

2. PROGRAM ANALYSIS

The Program Analysis of the College Course Registration and Subject Enrollment System focuses on understanding the functional requirements, system behavior, and interactions between different components. This analysis helps in designing an efficient and reliable system that meets the needs of students, administrators, and faculty members.

The system primarily consists of three main user modules: **Student**, **Administrator**, and **Faculty**. The student module allows users to log in, view available courses, check eligibility, and enroll in subjects. The administrator module is responsible for managing courses, student data, faculty assignments, and system configurations. The faculty module enables instructors to view assigned courses and student lists. Each module interacts with a centralized database where all information related to users, courses, and enrollments is stored and managed.

The input to the system includes student details, course information, faculty data, and registration requests. The system processes this data by validating user credentials, checking prerequisites, verifying seat availability, and ensuring there are no timetable conflicts. Once the validation is successful, the system updates the database and confirms the enrollment. In case of any issues, such as full capacity or unmet prerequisites, the system provides appropriate error messages.

The output of the system includes confirmation of course registration, updated course lists, enrollment status, and administrative reports. Reports generated by the system help in analyzing course demand, student participation, and resource allocation. These outputs are presented in a user-friendly format for easy understanding and decision-making.

From a technical perspective, the system can be developed using programming languages like Java and integrated with a database such as MySQL. It follows a structured design approach, often using concepts like modular programming and object-oriented design. This ensures that the system is maintainable, scalable, and easy to upgrade in the future.In conclusion, the program analysis ensures that all system requirements are clearly defined and properly implemented. It plays a crucial role in developing a robust, efficient, and error-free course registration system that enhances the overall academic management process.

2.1 Problem Definition

In many colleges and universities, the process of course registration and subject enrollment is still carried out using manual or semi-automated methods. These traditional systems involve paperwork, long queues, and significant administrative effort, making the process time-consuming and inefficient. Students often face difficulties in selecting courses due to limited access to accurate and updated information about subject availability, prerequisites, and schedules.One of the major problems in the existing system is the occurrence of errors such as duplicate registrations, incorrect data entry, and timetable clashes. Since the process is handled manually or with minimal automation, there is a higher chance of mistakes that can affect students' academic progress. Additionally, managing large volumes of student and cours data

challenging for administrators, leading to delays and confusion during peak registration periods.

Another significant issue is the lack of real-time updates and transparency. Students may not be aware of seat availability or changes in course schedules, resulting in inconvenience and frustration. Administrators also find it difficult to track enrollment status, generate reports, and make informed decisions regarding course allocation and resource management.

Furthermore, the absence of a centralized and secure system can lead to data inconsistency and potential loss of important information. Without proper access control, sensitive data may also be at risk. These challenges highlight the need for a reliable, efficient, and automated system.

Therefore, the College Course Registration and Subject Enrollment System is proposed to address these problems by providing a digital platform that ensures accuracy, efficiency, transparency, and secure management of academic processes.

2.2 Project Overview

The College Course Registration and Subject Enrollment System is a software-based project designed to automate and manage the process of course selection and student enrollment in educational institutions. The project aims to replace traditional manual registration methods with an efficient, reliable, and user-friendly digital system. It provides a centralized platform where students, administrators, and faculty members can interact and perform their respective tasks بسهولة and accurately.

This project focuses on developing a system that allows students to register for courses online by viewing available subjects, checking prerequisites, and selecting courses based on their academic requirements. The system ensures that students can access real-time information regarding course availability, seat capacity, and schedules. It also includes validation mechanisms to prevent errors such as duplicate enrollment and timetable conflicts, thereby improving the accuracy of the registration process.

From the administrative perspective, the system provides tools to manage course details, assign faculty, monitor student enrollments, and maintain academic records. Administrators can easily add, update, or delete course information and generate reports to analyze student enrollment trends and course demand. Faculty members can access their assigned courses and view the list of enrolled students.

The project is typically implemented using technologies such as Java for application development and MySQL for database management. It follows a modular and scalable design approach, making it easy to maintain and upgrade in the future. Additional features like online fee payment, attendance tracking, and result management can be integrated as extensions of the system

Overall, the College Course Registration and Subject Enrollment System enhances the efficiency and transparency of academic operations. It reduces administrative workload, saves time, minimizes errors, and improves the overall experience for students and staff, making it an essential solution for modern educational institutions.

2.3 Scopes and Limitations

Scope

Automation of Registration Process
The system automates course registration and subject enrollment, reducing manual work.

- **Centralized Data Management**
  Stores all student, course, and faculty information in a single database for easy access.
- **Student Access**
  Students can view courses, check eligibility, and enroll online from anywhere.
- **Admin Control**
  Administrators can manage courses, student records, and faculty assignments efficiently.
- **Real-Time Updates**
  Provides up-to-date information on course availability, schedules, and seat capacity.
- **Error Reduction**
  Minimizes issues like duplicate registration and timetable clashes through validation.

Limitations

- Dependence on Internet/Technology
  Requires internet access and proper devices to use the system effectively.
- Initial Setup Cost
  Development and implementation may require time and financial investment.
- Technical Issues
  System failures, bugs, or server downtime can affect availability.
- Limited User Knowledge
  Users with low technical skills may find it difficult initially.
- Data Security Risks
  Although secure, the system may still be vulnerable to cyber threats if not properly maintained.

3. SYSTEM DESIGN

The System Design of the College Course Registration and Subject Enrollment System defines the overall structure, components, and workflow of the application. It focuses on how different modules interact with each other and how data flows within the system to ensure efficient and reliable operation.

The system follows a **three-tier architecture**, which includes the **Presentation Layer**, **Application Layer**, and **Database Layer**. The Presentation Layer (user interface) allows users such as students, administrators, and faculty to interact with the system through login pages, dashboards, and forms.

3.1 Modules

1. Student Module

- Student registration and login
- View available courses and subject details
- Check prerequisites and eligibility
- Enroll in courses/subjects
- View enrolled courses and timetable
- Track registration status

2. Admin Module

- Admin login and authentication
- Add, update, and delete courses
- Manage student records
- Assign faculty to subjects
- Set course capacity and schedules
- Monitor student enrollments
- Generate reports

3. Faculty Module

- Faculty login
- View assigned subjects
- Access list of enrolled students
- Manage course-related information

4. Course Management Module

- Create and manage course details
- Define prerequisites and credits
- Update course schedules and timings

5. Enrollment Module

- Handle student course registration
- Validate eligibility and prerequisites
- Prevent duplicate registration
- Manage add/drop of courses

6. Authentication Module

- Secure login system for all users
- Role-based access control (Student/Admin/Faculty)
- Password management and security

7. Report Generation Module

- Generate reports on student enrollment
- Course-wise and department-wise reports
- Academic performance summaries

8. Database Module

- Store student, course, and faculty data
- Maintain enrollment records
- Ensure data consistency and integrity

4. METHODOLOGY

The methodology used for developing the College Course Registration and Subject Enrollment System follows a structured approach to ensure the system is efficient, reliable, and user-friendly. The development process is carried out in several phases, each focusing on a specific aspect of the system.

Initially, the **requirement analysis phase** is conducted to understand the needs of students, administrators, and faculty members. In this phase, all functional and non-functional requirements are identified, such as course registration, data management, security, and report generation. This helps in defining the scope and objectives of the system clearly.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

4.1. Problem Analysis

Identify issues in the existing manual system such as time consumption, errors, and lack of transparency.

- Analyze user requirements (students, admin, faculty) and system needs.
- Define objectives and scope for developing an automated solution.

4.2. System Design

Design system architecture (modules like student, admin, faculty).

- Create database structure for storing student, course, and enrollment data.
- Plan user interface and data flow for smooth system operation.

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

Develop the system using programming languages like Java.

- Implement modules such as login, course registration, and report generation.
- Integrate the system with a database (e.g., MySQL).

4.4. Testing and Validation

Perform unit testing for individual modules.

- Conduct system testing to ensure all components work together properly.
- Validate outputs to ensure accuracy and error-free performance.

4.5. Result and Documentation

Provide successful implementation of an automated registration system.

- Document system features, user manual, and technical details.
- Maintain records for future reference and system improvements.

4.6. Algorithm

Algorithm for the **Course Registration Process** :

Algorithm: Course Registration Process

1. **Start**
2. Display login page
3. Enter username and password
4. Validate user credentials
    - If invalid → Display error message and go back to login

5. Display list of available courses

6. Student selects desired course(s)

7. For each selected course:
   a. Check eligibility (prerequisites)
   b. Check seat availability
   c. Check timetable for conflicts

8. If all conditions are satisfied:

   - Register the student for the course
   - Update database (reduce available seats)
   - Display “Enrollment Successful”

9. Else:

   - Display appropriate error message (e.g., “Seats Full” / “Not Eligible” / “Time Conflict”)

10. Allow student to view enrolled courses

11. Logout option

12. **End**

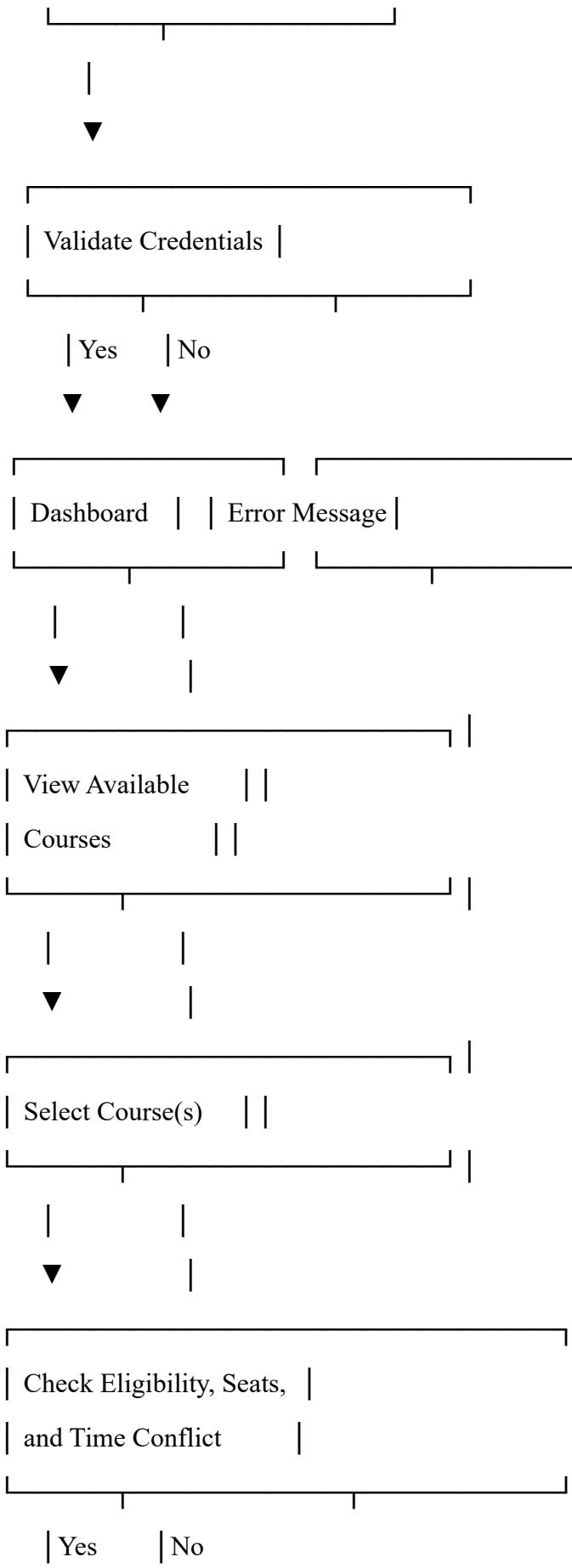
4.7. Flowchart

Students should draw a flowchart showing program flow.

Example flow:

```
┌──────────────┐
│   Start      │
└──────┬───────┘
       │
       ▼

┌──────────────────────┐
│ Enter Login          │
│ Credentials          │
```


Validate Credentials
Yes
No
Dashboard
Error Message
View Available Courses
Select Course(s)
Check Eligibility, Seats, and Time Conflict
Yes
No

```
  ▼       ▼
┌──────────────────┐  ┌──────────────────────┐
│ Register         │  │ Show Error           │
│ Course           │  │ Message              │
└──────┬───────────┘  └──────┬───────────────┘
   │          │
   ▼          │
┌──────────────────────────┐│
│ Update Database          │
└──────┬───────────────────┘│
   │          │
   ▼          │
┌──────────────────────────┐│
│ Show Confirmation        │
└──────┬───────────────────┘│
   │          │
   ▼          │
     ┌──────────────┐
     │  Logout      │
     └──────┬───────┘
        ▼
      ┌──────────────┐
      │   End        │
      └──────────────┘
```

5. SOURCE CODE

```
import java.util.*;

class Student {
    int id;
    String name;
    List<String> subjects;

    Student(int id, String name) {
        this.id = id;
        this.name = name;
        this.subjects = new ArrayList<>();
    }

    void enroll(String subject) {
        subjects.add(subject);
    }

    void display() {
        System.out.println("ID: " + id);
        System.out.println("Name: " + name);
        System.out.println("Subjects: " + subjects);
        System.out.println("----------------------");
    }
}

public class CourseRegistrationSystem{
    static ArrayList<Student> students = new ArrayList<>();
```

```
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    while (true) {
        System.out.println("\n1. Add Student");
        System.out.println("2. Enroll Subject");
        System.out.println("3. View Records");
        System.out.println("4. Exit");

        System.out.print("Enter choice: ");
        int ch = sc.nextInt();

        switch (ch) {
            case 1:
                System.out.print("Enter ID: ");
                int id = sc.nextInt();
                sc.nextLine();

                System.out.print("Enter Name: ");
                String name = sc.nextLine();

                students.add(new Student(id, name));
                System.out.println("Student added!");
                break;

            case 2:
                System.out.print("Enter Student ID: ");
                int sid = sc.nextInt();
                sc.nextLine();
```

```
                for (Student s : students) {
                    if (s.id == sid) {
                        System.out.print("Enter Subject: ");
                        String sub = sc.nextLine();
                        s.enroll(sub);
                        System.out.println("Subject enrolled!");
                    }
                }
                break;

            case 3:
                for (Student s : students) {
                    s.display();
                }
                break;

            case 4:
                System.exit(0);
            }
        }
    }
}
```

6. INPUT AND OUTPUT SCREENS

INPUT SCREEN

```
CourseRegistrationSystem (1) [Java Application] C:\Users\KGCAS\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.9.v20251105-0741\jre\bin\javaw.exe  (10-Apr

1. Add Student
2. Enroll Subject
3. View Records
4. Exit
Enter choice:
```

OUTPUT SCREEN

```
CourseRegistrationSystem (1) [Java Application] C:\Users\KGCAS\.p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.9.v20251105-0741\jre\bin\javaw.exe  (10-Apr

1. Add Student
2. Enroll Subject
3. View Records
4. Exit
Enter choice: 1
Enter ID: 102
Enter Name: ajay
Student added!

1. Add Student
2. Enroll Subject
3. View Records
4. Exit
Enter choice: 2
Enter Student ID: 102
Enter Subject: java maths
Subject enrolled!

1. Add Student
2. Enroll Subject
3. View Records
4. Exit
Enter choice: 3
ID: 102
Name: ajay
Subjects: [java maths]
---------------------
```

7. DISCUSSION & CONCLUSION

7.1 Discussion

The developed system demonstrates how a simple program can effectively manage student registration and subject enrollment using basic programming concepts such as classes, objects, lists, and user input handling. The implementation provides a menu-driven interface where users can add student details, enroll subjects, and view records. This approach makes the system easy to understand and use, especially for beginners learning Java programming.

The use of an Array list to store student records allows dynamic data handling without a fixed size limitation. Each student object maintains its own list of enrolled subjects, which helps in organizing data efficiently. The program successfully handles multiple students and allows subject enrollment based on student ID, showing a clear application of object-oriented programming principles like encapsulation and modularity.

However, the system has some limitations. It does not include validation for duplicate student IDs or invalid inputs. Additionally, the data is stored only temporarily in memory, meaning all records are lost when the program is closed. Despite these limitations, the system serves as a strong foundation and can be enhanced further by integrating a database, adding a graphical user interface, and improving error handling.

7.2 Conclusion

The College Course Registration and Subject Enrollment System successfully achieves its objective of automating the basic process of student registration and subject enrollment. It reduces manual effort, improves data organization, and provides a simple and efficient way to manage student records.

The project highlights the practical use of Java programming concepts in solving real-world problems. It also demonstrates how a structured approach can improve the efficiency of academic processes. Although the current system is basic, it can be further developed into a full-fledged application with advanced features like database integration, user authentication, and web-based access.

In conclusion, this project is a useful step toward understanding software development for academic management systems and provides a solid base for future enhancements and real-time

8. REFERENCES

The following resources were used during the development of this project:

Books

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- ***E. Balagurusamy***, *Programming with Java, McGraw-Hill Education.*
- ***Cay S. Horstmann***, *Core Java Volume I – Fundamentals, Pearson Education.*

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